RC CAR (CHASSIS)

PROJECT NAME:	Design And Development Of RC Car
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WHY THIS PROJECT ?

- To gain practical knowledge of RC vehicle mechanics, electronics, and control systems.
- To prepare for RC competitions, testing speed, stability, and handling.
- To understand component interaction, troubleshooting, and real-te system performance.
- To enhance engineering skills like problem-solving, teamwork, and system integration.

INTRODUCTION

- Used a ready-made RC chassis frame as the base structure.
- Mounted and integrated all components like motors, electronics, battery, and drivetrain ourselves.
- Learned how mechanical and electronic systems work together in an RC car.
- Gained hands-on experience in assembly, testing, and troubleshooting.
- Prepared for competitions by ensuring performance, stability, and control.

KEY FEATURES

- Ready-made nylon chassis frame with roll cage and front bumper for durability.
- Four-link suspension with oil-filled shock absorbers for smooth handling.
- Two-speed transmission and high-torque motor for versatile performance.
- Metal-geared differentials and universal drive shafts for efficient power transfer.
- ESC, servos, LiPo battery, and 2.4GHz transmitter for precise electronic control.

ARCHITECTURE

- Metallic/Nylon Chassis Frame: Base structure that supports and holds all components together.
- Motor & Transmission System: High-torque motor and two-speed transmission provide motion and power to the wheels.
- Drive System: Differentials, drive shafts, and portal axles transfer power efficiently to the wheels.
- Control Electronics: ESC, receiver, and servos control speed, steering, and gear shifting.
- Power System: LiPo battery and power distribution provide electrical energy to all components.

RESULTS

- Successfully mounted and integrated all components on the ready-made chassis.
- RC car performed smoothly with good control and stability.
- Gained practical experience in assembly, testing, and troubleshooting electronics and mechanical parts.
- Prepared for RC competitions, understanding how design and component choices affect real-world performance.

FUTURE SCOPE

1. **Refine design to improve handling and stability in competitions.**
2. **se lightweight composites or alloys for improved performance.**
3. • **Upgrade motors, ESC, or battery for higher speed and efficiency.**